

Program:

```
import nltk

from nltk.tag import hmm

from nltk.corpus import treebank

# Download required NLTK resources

nltk.download('treebank')

nltk.download('universal_tagset')

# Load a portion of the Penn Treebank corpus

train_data = treebank.tagged_sents(tagset='universal')[:3000]

test_data = treebank.tagged_sents(tagset='universal')[3000:]

# Train an HMM model

trainer = hmm.HiddenMarkovModelTrainer()

hmm_tagger = trainer.train(train_data)

# Test the model

accuracy = hmm_tagger.evaluate(test_data)

print(f'HMM Tagger Accuracy: {accuracy:.4f}')

# Test with a sentence

sentence = "The quick brown fox jumps over the lazy dog".split()

print("Tagged Sentence:")

print(hmm_tagger.tag(sentence))
```